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# Greedy Algorithm for Multi-Armed Bandit
# Using Arithmetic Mean to estimate Q(a)

# Number of machines
n = int(input("Enter number of machines: "))

machines = {}

# Input rewards for each machine
for i in range(n):
    name = input(f"Enter machine name {i + 1}: ")
    rewards = list(map(int, input("Enter observed rewards: ").split()))
    machines[name] = rewards

# Calculate Q(a) for each machine
q_values = {}

for machine, rewards in machines.items():
    q = sum(rewards) / len(rewards)
    q_values[machine] = q

# Display estimated Q(a)
print("\nEstimated Q(a) values:")

for machine, rewards in machines.items():
    print(f"\nMachine {machine}")
    print(f"Rewards: {rewards}")
    print(f"Q({machine}) = {q_values[machine]:.2f}")

# Greedy selection
selected_machine = max(q_values, key=q_values.get)

print("\nGreedy Selected Machine:", selected_machine)
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```
Enter number of machines: 3
Enter machine name 1: A
Enter observed rewards: 3 2 4
Enter machine name 2: B
Enter observed rewards: 6 7
Enter machine name 3: C
Enter observed rewards: 5 4 6
```

Estimated Q(a) values:

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Machine A
Rewards: [3, 2, 4]
Q(A) = 3.00
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Machine B
Rewards: [6, 7]
 $Q(B) = 6.50$

Machine C
Rewards: [5, 4, 6]
 $Q(C) = 5.00$

Greedy Selected Machine: B